

Kai Barker

Open to relocation | (530)-575-7568 | kaiandreasbarker@gmail.com | [linkedin.com/in/kaiabarker](https://www.linkedin.com/in/kaiabarker) | kai-portfolio-website.vercel.app

EDUCATION

University of California, Santa Barbara

Bachelor of Science (B.S.), Statistics and Data Science

September 2022 - June 2026

Santa Barbara, CA

Major GPA: 3.90/4.0

Activities: Consult Your Community, Statistics & Data Science Club, Data Science Collaborative, Ski & Snow Club

Coursework: Linear Algebra, Optimization, Machine Learning, Design of Experiments, Time Series, Adv. Statistical Models, Big Data

SKILLS & CERTIFICATES

Languages & Tools: SQL, Python (pandas, NumPy), R (tidyverse, dplyr), Microsoft Excel, PostgreSQL, Git, LaTeX, API Integration, HTML/CSS

Big Data & Engineering: Apache Spark, PySpark, GCP (BigQuery, Dataproc, GCS), ETL Pipelines, Distributed SQL, Data Modeling

Analytics & Visualization: Tableau, Plotly, Seaborn, Matplotlib, ggplot2, Hypothesis Testing, PowerPoint, Interactive Dashboards, KPIs

Machine Learning & Statistics: Regression, Scikit-Learn, XGBoost, PCA, Feature Selection (LASSO, Ridge), Bootstrap/Permutation Testing

Certificates: HackerRank SQL (2026), Anthropic AI Fluency Framework & Foundations (2026), JP Morgan Chase SWE (2025)

EXPERIENCE

Data and Systems Consultant

June 2026 - Present

Santa Barbara Wildlife Care Network

Santa Barbara, CA

- Identified 16-20 hours of weekly coordination overhead and \$1,000+/year in software costs through volunteer experience and stakeholder interviews; engaged as outside consultant to design and build a replacement system for 250+ volunteers.
- Built a PostgreSQL backend and React frontend with automated check-in, shift management, and onboarding workflows, projected to reduce manual coordination overhead to near zero and cut software costs to under \$14/month.

Engagement Manager and Business Analyst

October 2025 - June 2026

Consult Your Community

Santa Barbara, CA

- Led a 16-week EdTech market-entry engagement as Engagement Manager, directing a 7-person team across 3 workstreams toward the client's first paid B2B pilot.
- Modeled \$26M in recoverable funding across 115 CA community colleges by building an Excel financial model and Tableau dashboards, communicating findings directly to non-technical clients to drive recommendations.
- Translated data findings, including that 70% of corporate clients had lapsed within a year, into actionable deliverables: a CRM system, automated email reactivation campaigns, and a buyer-ready SOP for acquisition preparation.
- Identified \$40,000 in recoverable annual revenue by extracting and merging transactional data from Shopify, Square, and client records, then segmenting corporate accounts by order value, activity, and lifetime value.

Data Science Researcher

September 2025 - June 2026

Department of Statistics and Applied Probability | Advised by Department Chair

Santa Barbara, CA

- Investigated whether congressional stock trades carry predictive signal, building an end-to-end Python pipeline to clean and reconcile 137,000+ disclosure records into 15,000+ purchase events and measure CAPM-adjusted abnormal returns against the S&P 500.
- Identified a statistically significant signal in a subset of trades, top-performing members and coordinated purchases generated abnormal 30-day returns of +1.4% while unfiltered trades showed none, distinguishing a genuine signal from a tradable strategy.
- Authored a research paper and presented findings to the department chair, applying Random Matrix Theory covariance denoising and convex portfolio optimization, and documenting results with full visualizations and a limitations analysis.

Research Assistant

September 2024 - June 2026

Foundations of Financial Technology Lab

Santa Barbara, CA

- Quantified extreme wealth concentration in Ethereum (Gini coefficient of 0.96) by building a pipeline to process 20M+ blocks and 100M+ wallets across a 5-year period, showing the top 1% of accounts control 95%+ of Ether supply.
- Improved accuracy of distribution metrics by filtering 65,000+ institutional and artificial accounts to isolate individual holdings.
- Credited for analysis contributions to a working blockchain inequality paper alongside researchers from Yale, Cornell, and UCSB.

PROJECTS

Gene Therapy Off-Target Effect Analysis

May 2026

- Engineered an end-to-end ETL pipeline on GCP using PySpark, Dataproc, and BigQuery SQL to process a 78,893-gene expression matrix and 705,000+ disease association records across 22 candidate gene therapy treatments for Angelman syndrome research.
- Determined the single safest of 22 candidate treatments by building a distributed validation framework that ran 1,800+ model fits in parallel and flagged statistically significant off-target effects ($p < 0.001$).

NeurIPS Ariel Data Challenge

July 2025

- Designed a signal processing pipeline to calibrate 200+ GB of raw telescope data across 14,000 files and 1,100+ exoplanets, correcting detector noise and instrument distortion to recover usable atmospheric signal for the European Space Agency's Ariel mission.
- Constructed a weighted ensemble of Gaussian Process Regression, XGBoost, and Ridge models to predict atmospheric composition across 283 wavelength channels, tuning uncertainty estimates via grid search to optimize Gaussian Log Likelihood.